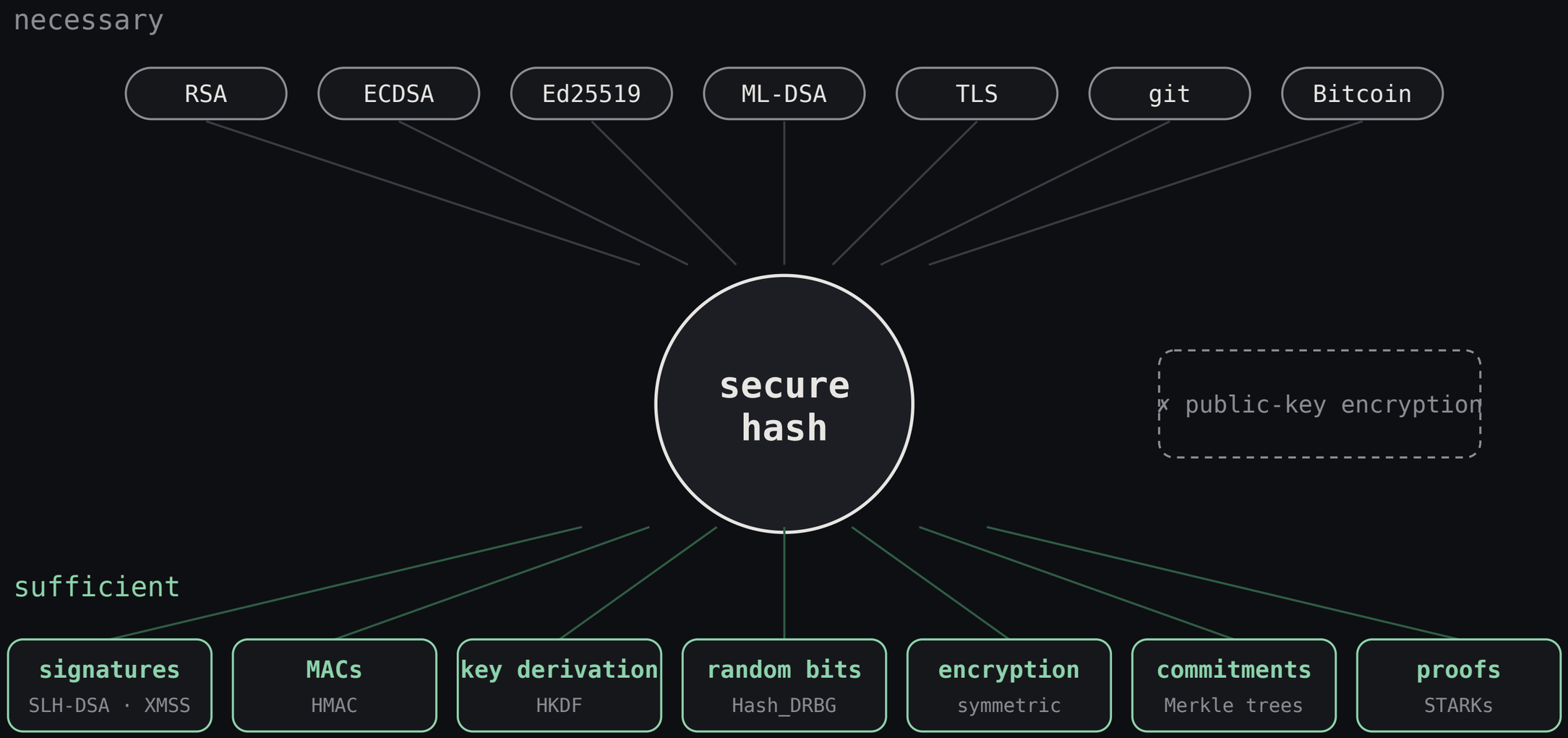
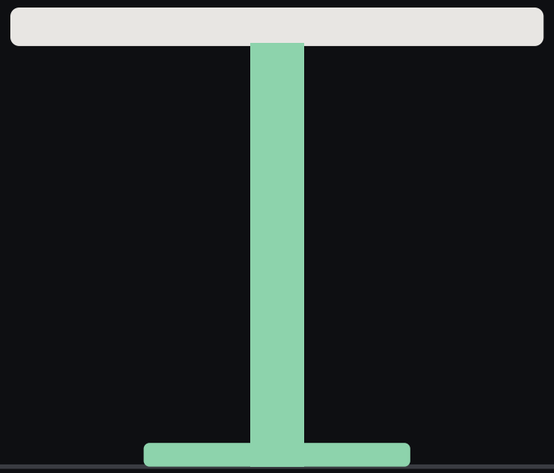


https://HashSma.sh


necessary
RSA
ECDSA
Ed25519
ML-DSA
TLS
git
Bitcoin
secure hash
✗ public-key encryption
sufficient
signatures
SLH-DSA · XMSS
MACs
HMAC
key derivation
HKDF
random bits
Hash_DRBG
encryption
symmetric
commitments
Merkle trees
proofs
STARKs

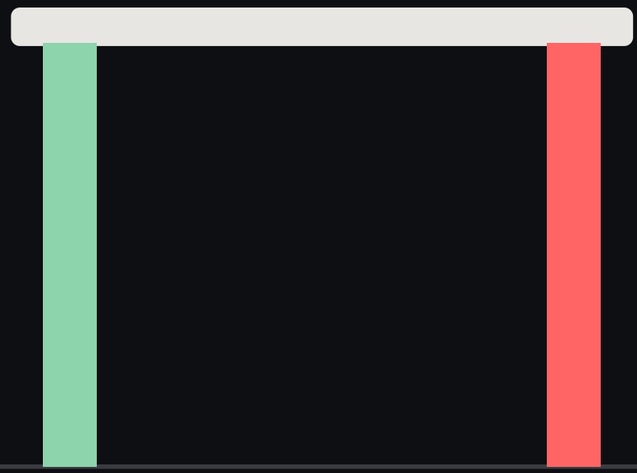
hash-based system



hash

falls if the hash breaks

anything else

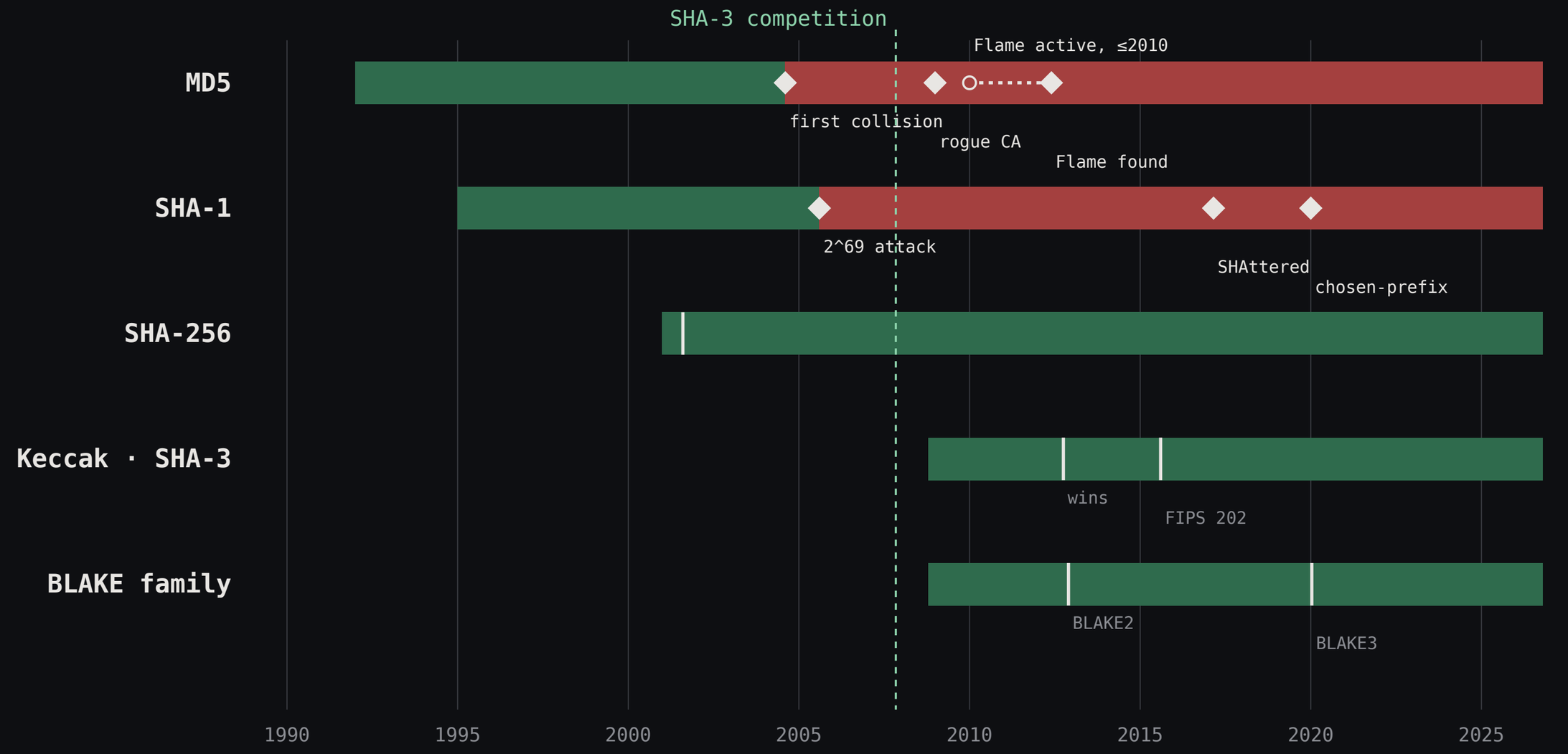


hash **lattices, curves, McEliece, etc.**

falls if the hash breaks
or if the other leg does

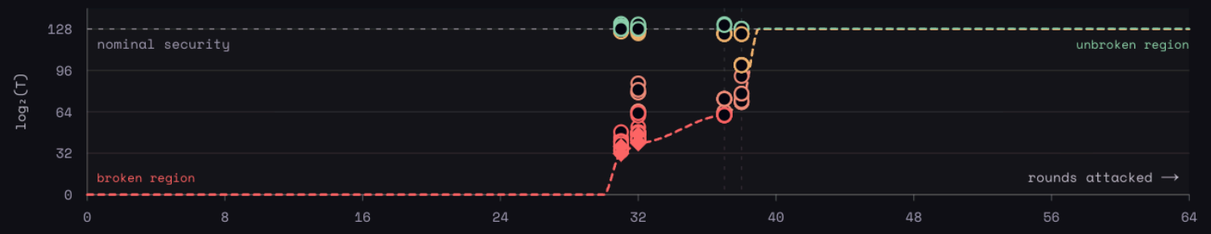


After Zooko, “lessons from the history of attacks on secure hash functions” (2017), Figure 0



green: no collision attack · red: collisions in practice · ◆ collision milestone

SHA-256 171 submissions View Results



SHA3-256 119 submissions



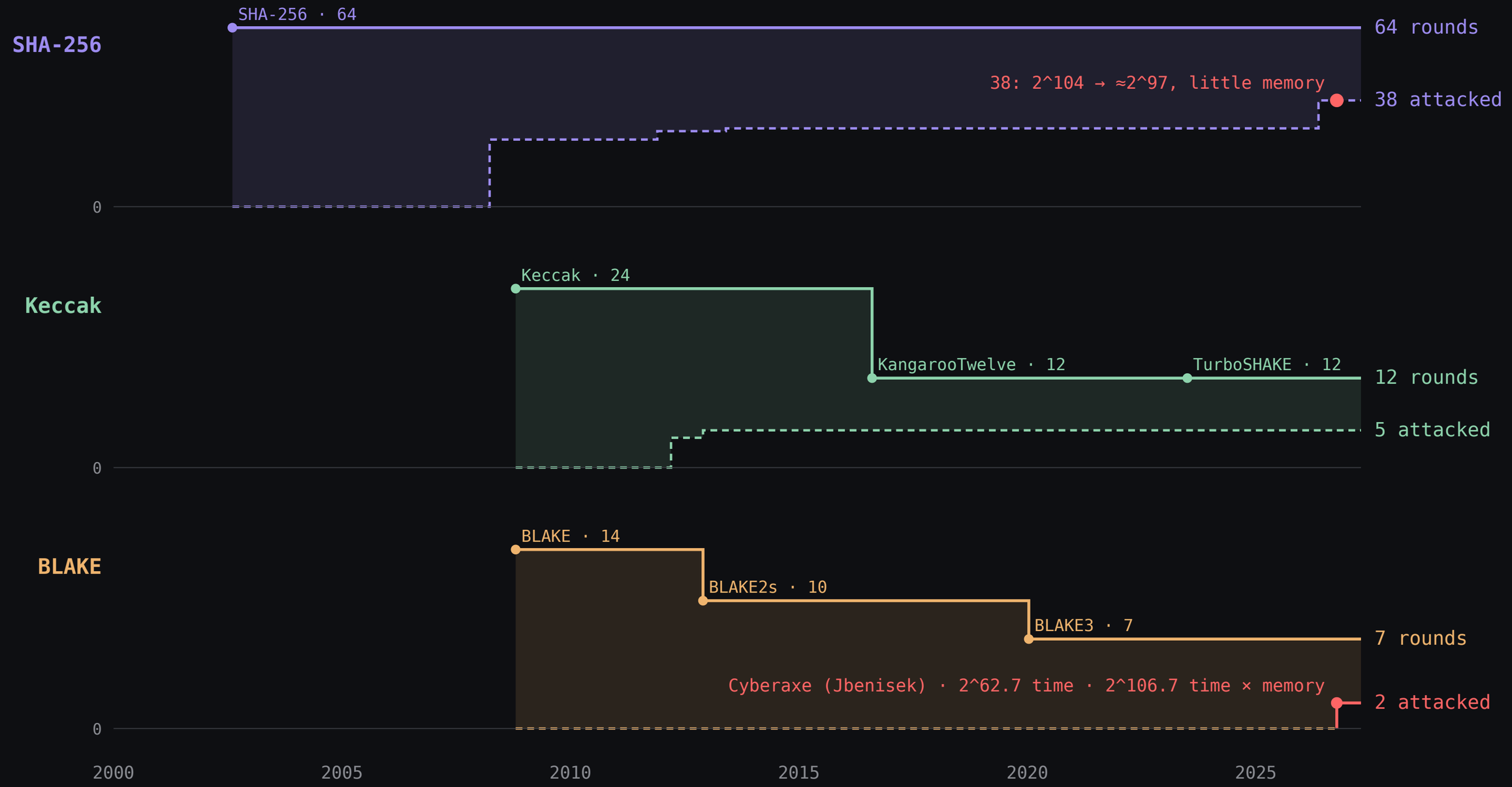
BLAKE3 133 submissions



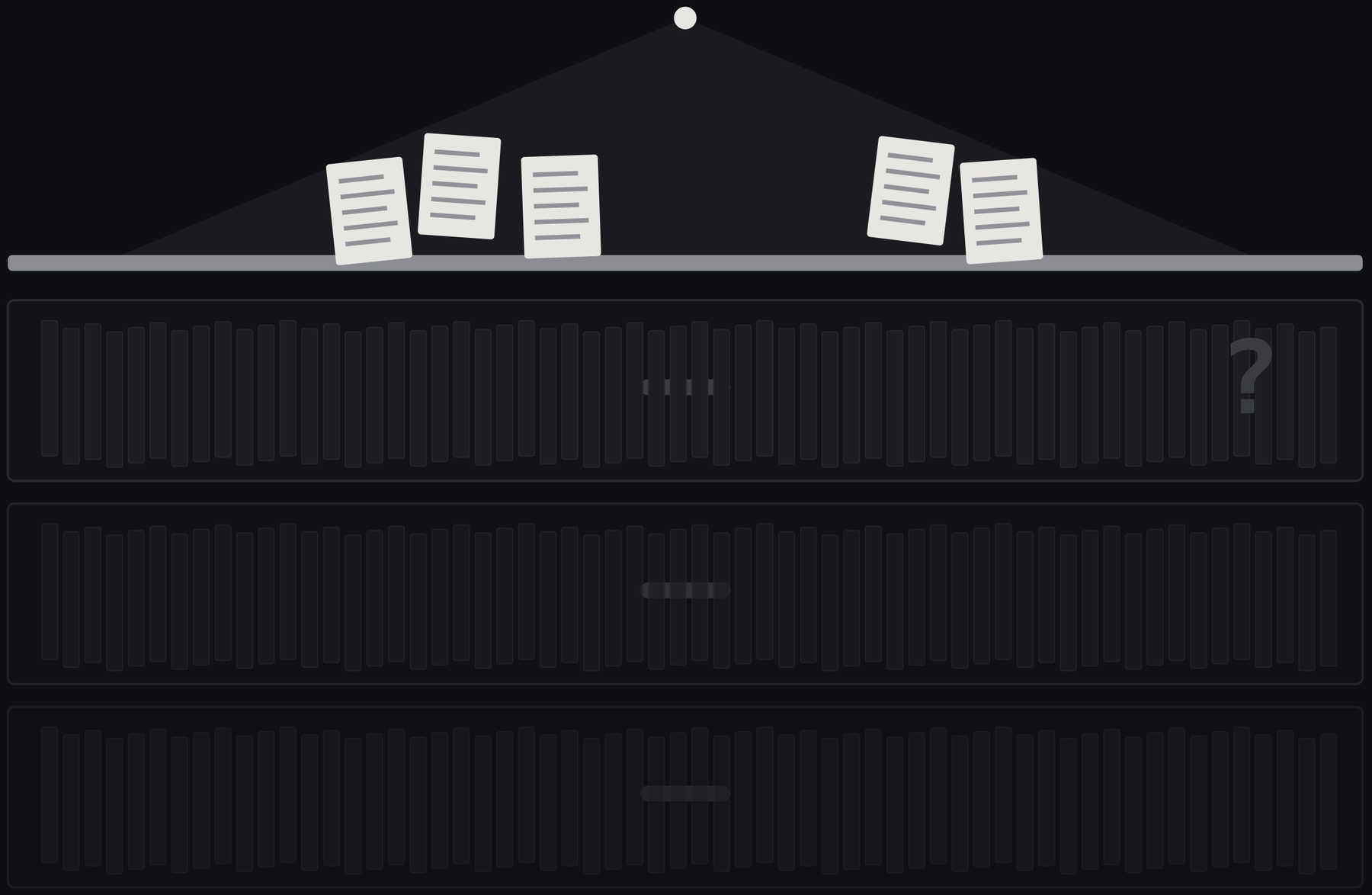
Poseidon 0 submissions

Coming Soon

In review Accepted Tentative frontier Safe Wounded Unsafe VERY Unsafe



solid: rounds the designers chose · dashed: rounds with a published full collision · shaded: the margin · red: HashSmash





Questions

https://HashSma.sh

https://x.com/shieldedlabs

https://x.com/zooko